Mathematics in Java - Study Guide

1. Mathematical Operations in Java

Basic Arithmetic Operators

Java provides several arithmetic operators for mathematical calculations:

- `+` Addition
- `-` Subtraction
- `*` Multiplication
- `/` Division
- `%` Modulus (remainder)

```java
int a = 15, b = 4;
System.out.println("Addition: " + (a + b));        // 19
System.out.println("Subtraction: " + (a - b));     // 11
System.out.println("Multiplication: " + (a * b));  // 60
System.out.println("Division: " + (a / b));        // 3
System.out.println("Modulus: " + (a % b));         // 3
```

Integer vs Floating-Point Division

```java
int x = 7, y = 2;
double result1 = x / y;           // 3.0 (integer division, then cast)
double result2 = (double)x / y;   // 3.5 (floating-point division)
double result3 = x / 2.0;         // 3.5 (floating-point division)
```

2. Math Class Methods

Common Math Functions

```java
import java.lang.Math;

// Power and roots
double power = Math.pow(2, 3);      // 8.0 (2^3)
double sqrt = Math.sqrt(16);        // 4.0
double cbrt = Math.cbrt(27);        // 3.0 (cube root)

// Absolute value
int abs1 = Math.abs(-10);           // 10
double abs2 = Math.abs(-3.14);      // 3.14

// Min and Max
int min = Math.min(5, 3);           // 3
int max = Math.max(5, 3);           // 5

// Rounding functions
double ceil = Math.ceil(4.3);       // 5.0
double floor = Math.floor(4.7);     // 4.0
long round = Math.round(4.6);       // 5
```

Trigonometric Functions

```java
// Angles in radians
double sin = Math.sin(Math.PI / 2);     // 1.0
double cos = Math.cos(0);               // 1.0
double tan = Math.tan(Math.PI / 4);     // 1.0

// Convert degrees to radians
double radians = Math.toRadians(90);    // π/2
double degrees = Math.toDegrees(Math.PI); // 180.0
```

Logarithmic and Exponential Functions

```java
double log = Math.log(Math.E);          // 1.0 (natural log)
double log10 = Math.log10(100);         // 2.0 (base-10 log)
double exp = Math.exp(1);               // 2.718... (e^1)
```

3. Advanced Mathematical Problems

Prime Number Checking

```java
public static boolean isPrime(int n) {
    if (n <= 1) return false;
    if (n <= 3) return true;
    if (n % 2 == 0 || n % 3 == 0) return false;

    for (int i = 5; i * i <= n; i += 6) {
        if (n % i == 0 || n % (i + 2) == 0) {
            return false;
        }
    }
    return true;
}
```

Factorial Calculation

```java
public static long factorial(int n) {
    if (n <= 1) return 1;
    long result = 1;
    for (int i = 2; i <= n; i++) {
        result *= i;
    }
    return result;
}

// Recursive approach
public static long factorialRecursive(int n) {
    if (n <= 1) return 1;
    return n * factorialRecursive(n - 1);
}
```

Fibonacci Sequence

```java
public static int fibonacci(int n) {
    if (n <= 1) return n;

    int prev1 = 0, prev2 = 1;
    for (int i = 2; i <= n; i++) {
        int current = prev1 + prev2;
        prev1 = prev2;
        prev2 = current;
    }
    return prev2;
}
```

Greatest Common Divisor (GCD)

```java
public static int gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

// Recursive approach
public static int gcdRecursive(int a, int b) {
    if (b == 0) return a;
    return gcdRecursive(b, a % b);
}
```

Least Common Multiple (LCM)

```java
public static int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}
```

4. Number System Conversions

Binary, Octal, and Hexadecimal

```java
int decimal = 42;

// Convert to different bases
String binary = Integer.toBinaryString(decimal);    // "101010"
String octal = Integer.toOctalString(decimal);      // "52"
String hex = Integer.toHexString(decimal);          // "2a"

// Parse from different bases
int fromBinary = Integer.parseInt("101010", 2);     // 42
int fromOctal = Integer.parseInt("52", 8);          // 42
int fromHex = Integer.parseInt("2a", 16);           // 42
```

5. Random Number Generation

Using Math.random()

```java
// Random double between 0.0 and 1.0
double random1 = Math.random();

// Random integer between 0 and 9
int random2 = (int)(Math.random() * 10);

// Random integer between min and max (inclusive)
public static int randomInRange(int min, int max) {
    return (int)(Math.random() * (max - min + 1)) + min;
}
```

Using Random Class

```java
import java.util.Random;

Random rand = new Random();
int randomInt = rand.nextInt(100);        // 0 to 99
double randomDouble = rand.nextDouble();   // 0.0 to 1.0
boolean randomBoolean = rand.nextBoolean();
```

6. Complex Mathematical Calculations

Quadratic Formula

```java
public static void quadraticFormula(double a, double b, double c) {
    double discriminant = b * b - 4 * a * c;

    if (discriminant > 0) {
        double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
        double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);
        System.out.println("Two real roots: " + root1 + " and " + root2);
    } else if (discriminant == 0) {
        double root = -b / (2 * a);
        System.out.println("One real root: " + root);
    } else {
        System.out.println("No real roots");
    }
}
```

Distance Between Two Points

```java
public static double distance(double x1, double y1, double x2, double y2) {
    return Math.sqrt(Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2));
}
```

Area and Perimeter Calculations

```java
// Circle
public static double circleArea(double radius) {
    return Math.PI * radius * radius;
}

public static double circlePerimeter(double radius) {
    return 2 * Math.PI * radius;
}

// Rectangle
public static double rectangleArea(double length, double width) {
    return length * width;
}

public static double rectanglePerimeter(double length, double width) {
    return 2 * (length + width);
}

// Triangle (using Heron's formula)
public static double triangleArea(double a, double b, double c) {
    double s = (a + b + c) / 2;
    return Math.sqrt(s * (s - a) * (s - b) * (s - c));
}
```

Quiz Questions

1. What is the result of `Math.pow(3, 4)` in Java? a) 12 b) 64 c) 81 d) 7

Correct Answer: c) 81

2. Which method would you use to find the smallest integer greater than or equal to 4.7? a) Math.floor(4.7) b) Math.ceil(4.7) c) Math.round(4.7) d) Math.abs(4.7)

Correct Answer: b) Math.ceil(4.7)

3. What does the expression `15 % 4` return? a) 3 b) 4 c) 3.75 d) 0

Correct Answer: a) 3

4. Which of the following correctly converts 90 degrees to radians? a) Math.toDegrees(90) b) Math.toRadians(90) c) Math.PI * 90 d) 90 / Math.PI

Correct Answer: b) Math.toRadians(90)

5. What is the time complexity of the optimized prime number checking algorithm shown in the study guide? a) O(n) b) O(n^2) c) O(√n) d) O(log n)

Correct Answer: c) O(√n)

Practice Coding Questions

1. Write a Java method that calculates the sum of all prime numbers between 1 and a given number n.

2. Create a program that generates the first n terms of the Fibonacci sequence and calculates their sum.

3. Write a method that determines if a given number is a perfect square without using Math.sqrt().

4. Implement a program that converts a decimal number to binary representation without using built-in methods like Integer.toBinaryString().

5. Create a method that calculates the digital root of a number (keep adding digits until you get a single digit). For example, digital root of 9875 is 2 (9+8+7+5=29, 2+9=11, 1+1=2).